

Arunachalam S

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SUMMARY

Professional with **2+ years** of experience in manufacturing, process improvement, and project coordination within the composite manufacturing industry. Skilled in **lean manufacturing, PFMEA, process optimization, tool validation**, and practical engineering solution development. Proficient in **ANSYS, SpaceClaim, SolidWorks, AutoCAD, Power BI, and Advanced Excel**. Experienced in cross-functional collaboration, continuous improvement, and data-driven problem solving, with a strong interest in leveraging engineering insights and analytics to support strategic decision-making and operational excellence.

TECHNICAL SKILLS

Operations: Lean Manufacturing, Process Optimization, PFMEA, Continuous Improvement, Root Cause Analysis, Process Critical Tool Validation, Kaizen, 5S, Inventory Management

Project and Product Lifecycle: NPI Process, Engineering Change Management, Project Management

Business and Analytics: Data-Driven Decision making, **Power BI, Advanced Excel**, Microsoft Office, Manufacturing Analytics

Technical and Engineering Tools: Ansys Workbench(FEA), SpaceClaim, SolidWorks, AutoCAD, MakerBot Replicator (3D Printing)

Programming Language: Python

EXPERIENCE

Senvion Wind Technology Pvt Ltd

Manufacturing Excellence Engineer

Jun 2024 – Present

Tiruchirappalli, Tamil Nadu

- Led a Curing Optimization Project that achieved a **71%** reduction in curing cycle time by standardizing processes and improving operational efficiency in alignment with industry standards.
- Utilized Microsoft Power BI for data-driven cycle time analysis, identifying bottlenecks and implementing corrective actions to optimize production throughput and reduce lead times.
- Initiated and executed a Paste Consumption Optimization Project, reducing paste consumption by 40 kg while maintaining product quality, process stability, and operational efficiency.
- Conducted Root Cause Analysis (RCA) and defect investigation activities to support the organization's Zero Defect manufacturing initiative and improve process reliability.
- Designed and implemented kitting trolleys using AutoCAD, streamlining material handling workflows and reducing logistics turnaround time by approximately **70%**

Bureau of Indian Standards

Research Analyst

May 2023 – Jul 2023

Delhi, India

- Contributed to the development of Indian standards for pneumatic bus door closure mechanisms by evaluating safety, reliability, and performance requirements.
- Conducted comparative analysis and industry benchmarking of pneumatic door systems through literature review and engineering data collection.
- Performed industrial visits to Ashok Leyland and ICAT to validate design practices and support the standardization framework development.

PROJECTS

Thermo-Mechanical Analysis of Carbon Fiber (CF) Reinforced PLA

- Fabricated and tested CF-reinforced PLA and neat PLA specimens using a MakerBot Replicator to evaluate mechanical and thermal properties across different infill patterns and densities, identifying superior thermal performance in CF-reinforced PLA compared to neat PLA.

Effect of Nanoparticles on CI Engine Characteristics

- Conducted performance, combustion, emission, noise, and vibration testing on a CI engine under varying fuel conditions using biodiesel blends with cerium oxide nanoparticles. Analyzed the effect of 12 biodiesel blends with different cerium oxide nanoparticle concentrations on engine characteristics, identifying improvements in overall efficiency and heat release rate, along with reductions in NOx emissions and engine noise.

EDUCATION & CERTIFICATIONS

NIT Tiruchirappalli

Bachelor of Technology in Mechanical Engineering (Minor in Management Studies)

Nov 2020 – May 2024

CGPA: 8.51

MTA: Introduction to Programming using Python (Microsoft)

2021

AWARDS & ACHIEVEMENTS

GATE AE AIR 582

2026

Senvion Project Completion Award

2026

Recognized for optimizing 11.5hrs of CT reduction in Shell Curing Process